

FULL-STACK REAL ESTATE MARKETPLACE

propmart

Project Documentation

Property discovery, listing, booking and
marketplace management.

Academic Project · 2026 · React · Express · MongoDB

PropMart | Project Documentation

Project Abstract

PropMart is a full-stack real-estate marketplace connecting buyers, sellers and administrators through a role-based property platform.

Buyers discover approved properties, save listings, schedule visits, contact sellers and publish reviews. Sellers submit and manage listings and handle buyer requests. Administrators manage users and verify property submissions.

OBJECTIVES

Buyer experience

- Search and filter properties.
- Save preferred listings.
- Book and track visits.
- Contact sellers and review properties.

Marketplace management

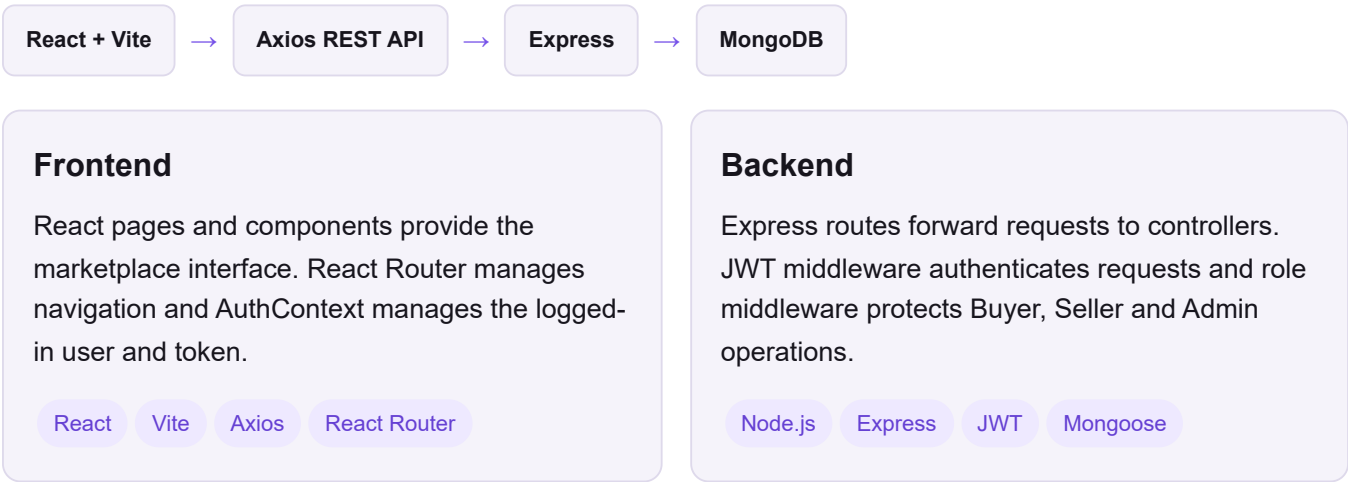
- Sellers manage property listings.
- Admins approve or reject submissions.
- Role-based access protects workflows.
- Booking status remains visible to both parties.

PROBLEM STATEMENT

Property discovery and seller coordination are often split across multiple tools. PropMart provides one workflow for discovery, verification, engagement and visit management.

System Architecture

The application separates the React client from the Express REST API and MongoDB data layer.



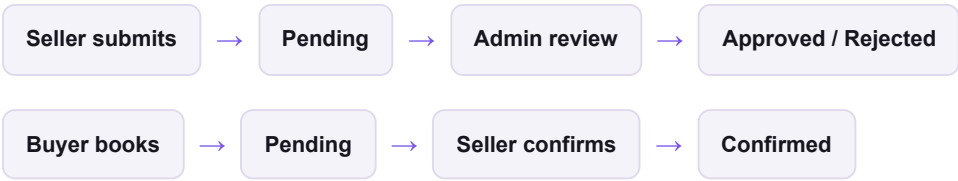
BACKEND LAYERS

Layer	Responsibility
Routes	Define API endpoints and apply middleware.
Controllers	Validate requests and execute business logic.
Models	Define MongoDB document structures.
Middleware	Verify tokens and authorize roles.

Functional Modules

Module	Functionality	Role
Authentication	Registration, login, bcrypt password hashing and JWT sessions.	All
Properties	Creation, editing, deletion, search, filters and details.	Buyer / Seller
Verification	Approve or reject seller listings before public discovery.	Admin
Wishlist	Add, view and remove saved approved properties.	Buyer
Bookings	Request visits, view status, confirm or cancel bookings.	Buyer / Seller
Inquiries	Send property questions to sellers.	Buyer
Reviews	Create, update and display ratings and comments.	Buyer
Administration	Dashboard statistics, users and property review tools.	Admin

CORE WORKFLOWS



Database and Usage

MongoDB stores User, Property, Booking, Wishlist, Review and Inquiry documents connected through ObjectId references.

ENVIRONMENT VARIABLES

```
PORT=5000  
MONGO_URI=your_mongodb_connection_string  
JWT_SECRET=your_private_jwt_secret
```

INSTALL AND RUN

```
npm install  
cd frontend  
npm install  
  
# Terminal 1 - project root  
npm run dev  
  
# Terminal 2 - frontend  
cd frontend  
npm run dev
```

Backend uses port 5000. Frontend uses the Vite URL shown in the terminal, normally port 5173.

VALIDATION

```
cd frontend  
npm run build  
npm run lint
```

VIVA DEMONSTRATION

1. Register Seller and Buyer accounts.
2. Submit and approve a property as Seller/Admin.
3. Browse it as Buyer and test filters and details.
4. Test wishlist save/remove and booking confirmation.
5. Send an inquiry and create or update a review.
6. Demonstrate Admin user and property management.

Testing and Future Scope

Functional checks

- Invalid credentials are rejected.
- Unauthorized roles are blocked.
- Only approved properties are public.
- Duplicate wishlist/reviews are handled.
- Loading, empty and error states are shown.

Presentation checks

- Responsive React interface.
- Black, off-white and lavender identity.
- Clear Buyer, Seller and Admin workflows.
- REST API separation from the UI.

FUTURE SCOPE

Seller inquiry management, image storage, notifications, advanced search, automated tests and production deployment can be added later.

CONCLUSION

PropMart demonstrates a complete full-stack marketplace workflow with authentication, role-based access, property verification, buyer engagement and seller operations.